

QUBO Formulation: Legal Document Extractive Summarization

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Problem Statement. Given a legal document partitioned into N candidate sentences, the goal is to select exactly K sentences to form an extractive summary. Each sentence $i \in \{0, 1, \dots, N-1\}$ is assigned an importance weight $w_i \geq 0$, and each unordered pair of distinct sentences (i, j) with $0 \leq i < j < N$ is assigned a redundancy penalty $p_{ij} \geq 0$ that applies if both sentences are simultaneously selected. The objective is to maximize the total summary score, defined as the sum of importance scores of the selected sentences minus the sum of redundancy penalties incurred between all selected pairs.

Decision Variables. For each candidate sentence $i \in \{0, 1, \dots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i equals 1 if sentence i is included in the extractive summary, and 0 otherwise. The complete solution is represented by the vector $x = (x_0, x_1, \dots, x_{N-1}) \in \{0, 1\}^N$.

QUBO Derivation. Step 1 — Original Objective and Sign Flip. The original problem is a maximization:

$$\max_{x \in \{0,1\}^N} \sum_{i=0}^{N-1} w_i x_i - \sum_{0 \leq i < j < N} p_{ij} x_i x_j. \quad (1)$$

Following QUBO convention, we convert this to a minimization problem by negating the objective:

$$\min_{x \in \{0,1\}^N} - \sum_{i=0}^{N-1} w_i x_i + \sum_{0 \leq i < j < N} p_{ij} x_i x_j. \quad (2)$$

Step 2 — Constraint Formulation. The requirement to select exactly K sentences is expressed as a cardinality constraint:

$$\sum_{i=0}^{N-1} x_i = K. \quad (3)$$

Step 3 — Quadratic Penalty Expansion. The constraint is enforced via a quadratic penalty term with coefficient $A > 0$:

$$A \left(\sum_{i=0}^{N-1} x_i - K \right)^2. \quad (4)$$

Expanding the square yields:

$$\left(\sum_{i=0}^{N-1} x_i - K \right)^2 = \left(\sum_{i=0}^{N-1} x_i \right)^2 - 2K \sum_{i=0}^{N-1} x_i + K^2. \quad (5)$$

Using the algebraic identity $\left(\sum_{i=0}^{N-1} x_i \right)^2 = \sum_{i=0}^{N-1} x_i^2 + 2 \sum_{0 \leq i < j < N} x_i x_j$ and the idempotent property of binary variables $x_i^2 = x_i$, we obtain:

$$\left(\sum_{i=0}^{N-1} x_i - K \right)^2 = \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j < N} x_i x_j - 2K \sum_{i=0}^{N-1} x_i + K^2. \quad (6)$$

Collecting linear terms gives the general penalty contribution:

$$A \left[(1 - 2K) \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j < N} x_i x_j + K^2 \right]. \quad (7)$$

Step 4 — Combined QUBO Hamiltonian. Adding the negated objective and the expanded penalty term yields the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} (A(1 - 2K) - w_i) x_i + \sum_{0 \leq i < j < N} (p_{ij} + 2A) x_i x_j + AK^2. \quad (8)$$

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from $H(x)$, the upper-triangular QUBO matrix entries $Q_{i,j}$ and constant offset c are given by:

$$Q_{i,i} = A(1 - 2K) - w_i, \quad \text{for } i \in \{0, \dots, N-1\}, \quad (9)$$

$$Q_{i,j} = p_{ij} + 2A, \quad \text{for } 0 \leq i < j < N, \quad (10)$$

$$Q_{i,j} = 0, \quad \text{for } i > j, \quad (11)$$

$$c = AK^2. \quad (12)$$

Penalty Weight Justification. The penalty coefficient A must be chosen large enough to ensure that any violation of the cardinality constraint incurs an energy cost strictly greater than the maximum possible improvement in the original objective function. The maximum change in the negated objective from flipping a single variable is bounded by $N \max_i |w_i| + N(N-1) \max_{i,j} |p_{ij}|$. Therefore, selecting $A > N \max_i |w_i| + N(N-1) \max_{i,j} |p_{ij}|$ guarantees that the quadratic penalty dominates any feasible objective gain, forcing the optimizer to prefer feasible solutions over infeasible ones.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^N$ recovers the optimal extractive summary because feasible assignments satisfying $\sum_{i=0}^{N-1} x_i = K$ incur zero penalty, meaning their energy exactly equals the negated summary score. Infeasible assignments violate the cardinality constraint and incur a penalty of at least A . By construction, A exceeds the maximum possible reduction in the objective function achievable by any constraint violation. Consequently, the global minimum of $H(x)$ cannot occur at an infeasible point, and must instead correspond to a feasible assignment that minimizes the negated objective, which is equivalent to maximizing the original summary score.

Illustrative Example. Consider a concrete instance with $N = 3$ candidate sentences and a target summary length of $K = 2$. Let the importance weights be $w_0 = 10$, $w_1 = 12$, $w_2 = 9$, and the redundancy penalties be $p_{01} = 3$, $p_{02} = 2$, $p_{12} = 4$. We set the penalty weight to $A = 100$.

The concrete maximization objective is:

$$\max 10x_0 + 12x_1 + 9x_2 - 3x_0x_1 - 2x_0x_2 - 4x_1x_2. \quad (13)$$

The concrete penalty term expands to:

$$100 [(1 - 4)(x_0 + x_1 + x_2) + 2(x_0x_1 + x_0x_2 + x_1x_2) + 4] = -300(x_0 + x_1 + x_2) + 200(x_0x_1 + x_0x_2 + x_1x_2) - \quad (14)$$

Combining these yields the concrete QUBO Hamiltonian:

$$H(x) = -310x_0 - 312x_1 - 309x_2 + 203x_0x_1 + 202x_0x_2 + 204x_1x_2 + 400. \quad (15)$$

For feasible solutions where $x_0 + x_1 + x_2 = 2$, the penalty contribution vanishes and $H(x)$ reduces to the negated objective. Evaluating the three feasible binary vectors:

$$H(1, 1, 0) = -19, \quad H(1, 0, 1) = -17, \quad H(0, 1, 1) = -17. \quad (16)$$

The global minimum is -19 at $x^* = (1, 1, 0)$. Thus, the optimal extractive summary consists of sentences 0 and 1, achieving a maximum summary score of 19.